

ML-DL-OPS LAB ASSIGNMENT - 4 REPORT

(Optimizing Transformer Translation with Ray Tune & Optuna)

Google Colab Link -

https://colab.research.google.com/drive/1M8qwvgyBLJurUq6FSrpud96aqxm_RHZk?usp=sharing

GitHub Link -

<https://github.com/Layaa-V/MLOps-LayaaVishwakarma-M25CSA017/tree/Assignment-4>

HuggingFace Model Link (for uploaded M25CSA017_ass_4_best_model.pth) -

<https://huggingface.co/Layaa-V/MLOPS-Assignment-4-en-to-hi/tree/main>

I. Baseline Metrics -

To establish a performance benchmark, the original static PyTorch training loop was executed without hyperparameter optimization. The model was trained to translate English to Hindi over a long-running cycle.

- Total Epochs : 100
- Time Taken : 3hrs 52mins
- Final Training Loss : 0.1156
- Baseline BLEU Score : 50.06

II. Hyperparameter Setup & Search Strategy -

To drastically improve training efficiency, the PyTorch training loop was refactored to interface with Ray Tune. The goal was to find a combination of parameters that could achieve a competitive BLEU score (approx. 50.0) in a fraction of the original 100 epochs.

Search Algorithm & Scheduler:

- Searcher: OptunaSearch was implemented to intelligently navigate the parameter space and predict the most promising configurations based on the validation loss.
- Scheduler: ASHAScheduler (Asynchronous Successive Halving Algorithm) was utilized with a grace period of 5 and a maximum of 10 epochs. This efficiently halted underperforming trials early, saving significant GPU compute time.

The Search Space:

Five key hyperparameters were selected for the tuning sweep to balance learning efficiency, model capacity, and regularization:

- Learning Rate (lr): Log-uniform distribution between 1e-5 and 1e-3.
- Batch Size (batch_size): Discrete choice of [16, 32, 64].

- Attention Heads (num_heads): Discrete choice of [4, 8] (ensuring d_model of 512 remains divisible).
- FeedForward Dimension (d_ff): Discrete choice of [1024, 2048].
- Dropout Rate (dropout): Uniform distribution between 0.1 and 0.4.

III. The Best Configuration -

After running the Ray Tune sweep across 20 distinct combinations, Optuna identified the following configuration as the optimal setup for minimizing translation loss:

- Learning Rate : 0.00014180047371283383
- Batch Size : 16
- Attention Heads : 8
- FeedForward Dimension : 2048
- Dropout Rate : 0.10013392490681565}

IV. Final Results & The Efficiency Challenge

The best-performing model was loaded and evaluated against the validation dataset to test the ultimate efficiency challenge.

- Total Epochs : 10
- Time Taken : 2hrs 48mins
- Final Validation Loss : 0.4735088855750633
- Final BLEU Score : 50.13

Conclusion:

The hyperparameter optimization was a massive success. The tuned model achieved a score of 50.13 in just 10 epochs, as compared to the older one achieving it in 100. This implemented a 90% reduction in training epochs compared to the 100-epoch baseline. This demonstrates the critical value of MLOps tools like Optuna in finding highly efficient, fast-converging model architectures.